Cryptography & Network Security
Assignment

Note: You need to write this assignment in file pages or in book

	Symmetric Encryption
1.	Use the Vigenere cipher with keyword "HEALTH" to encipher the message "Life is full of surprises".
2.	Use the Playfair cipher to encipher the message "**The key is hidden under the door pad**". The secret key can be made by filling the first and part of the second row with the word "**GUIDANCE**" and filling the rest of the matrix with the rest of the alphabet.
3.	Use a hill cipher to encipher the message "we live in an insecure world". Use the following key k= 03 02 05 07
4.	Encrypt the message "the house is being sold tonight" using Vigenere cipher with key "dollars". Ignore the space between words. Decrypt the message to get the plaintext
5.	Apply Vigenere cipher: Encrypt the word **"explanation"** using the key **"leg"**
6.	Encrypt the following using play fair cipher using the keyword **GUIDANCE**. "SWARAJ IS MY BIRTH RIGHT". Use X as blank space.
7.	Late one evening, the security analyst, Alex, was reviewing the logs of a company's network. Suddenly, a mysterious set of encrypted messages appeared in the logs. The messages seemed to indicate a potential security breach. As Alex investigated further, two distinct patterns emerged. **Pattern A**: Several log entries showed unauthorized access attempts and suspicious activity. It seemed like someone was actively trying to break into the system. **Pattern B:** In another set of logs, Alex noticed subtle changes in the network traffic, but no explicit signs of intrusion attempts. It was as if someone was silently observing the communication without leaving any traces. Alex knew that both active and passive attacks could pose serious threats. Can you identify which pattern represents an active attack and which one represents a passive attack? Provide a brief explanation for your choice.

8.	In a top-secret government agency, Agent X was assigned the task of safeguarding a highly classified database containing sensitive information. One day, a series of unusual events unfolded, and Agent X found himself facing challenges related to the core principles of information security **Scenario 1:** Agent X discovered that several classified documents had been leaked to unauthorized personnel. The information contained in these documents could compromise national security. Question: Which information security principle is primarily at risk in Scenario 1? Explain your choice. **Scenario 2:** Upon further investigation, Agent X realized that some of the mission plans stored in the database had been tampered with. The altered plans could lead to disastrous consequences if executed as is. Question: Which information security principle is primarily at risk in Scenario 2? Explain your choice. **Scenario 3:** During routine maintenance, the classified database experienced an unexpected downtime, rendering it temporarily inaccessible to authorized personnel. Question: Which information security principle is primarily at risk in Scenario 3? Explain your choice.
	Asymmetric Encryption
1	Find all primitive roots of 11.
2	Given a=161 and b=28, find gcd(a, b) and values of *s* and *t* using Extended Euclidean algorithm.
3	Find the multiplicative inverse of 11 in Z_{26}. Also check multiplicative inverse of 11 exists or not.
4	Use fast modulo exponent to find value of 81^{232} mod 256.

5	Perform encryption and decryption using the RSA algorithm, for the following: a) p = 11; q = 13, e = 11; M = 7 b) p = 17; q = 31, e = 7; M = 2 Find the private key and Provide proper justification of public key.
6	Use Rabin Crypto with p=23 and q=7. Message is 24. Calculate Cipher text and perform decryption process.
7	Users A and B use the Diffie-Hellman key exchange technique with a common prime q=71 and List out 5 Primitive root and select any one. and a primitive root. 1) If user A has private key XA=5, what is A's public key YA? 2) If user B has private key XB=12, what is B's public key YB? 3) c. What is the shared secret key?
8	Consider an ELGamal scheme with a common prime q=71 and a primitive rootα=7. If B has public key YB = 3 and A chose the random integer k=2, what is the ciphertext of M=30? b. If A now chooses a different value of k so that the encoding of M=30 is C=(59,C2), what is the integer C2?